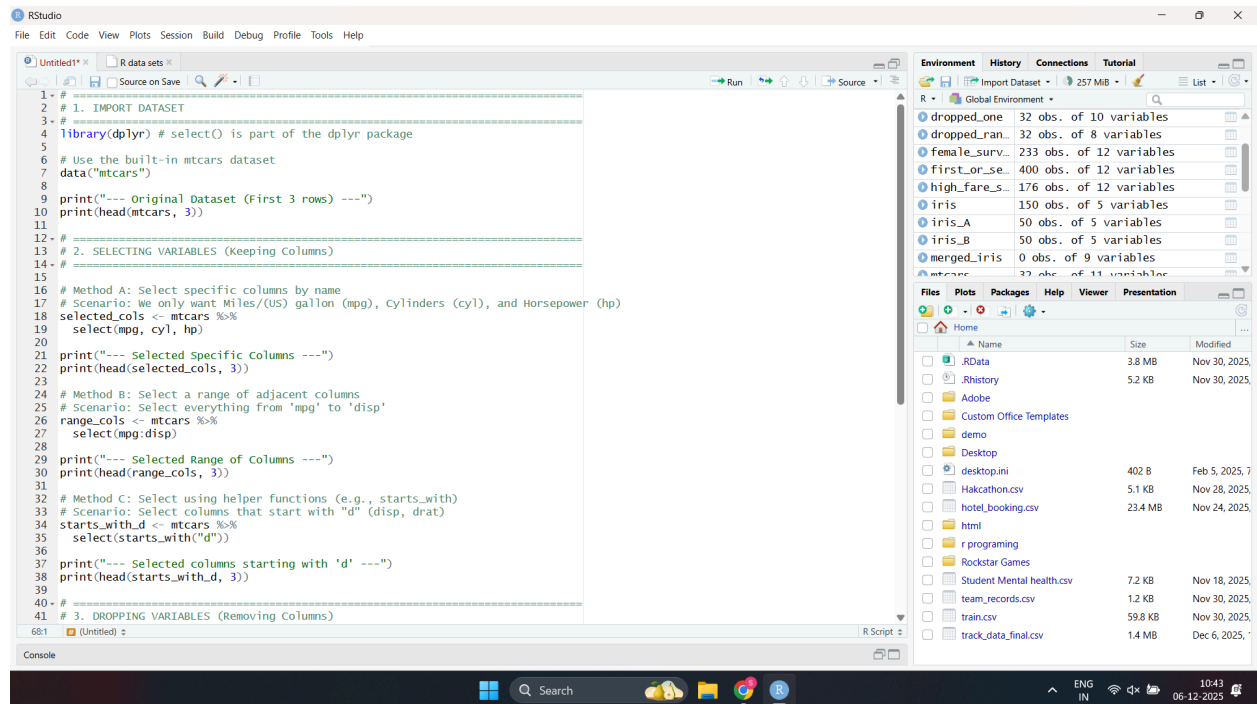


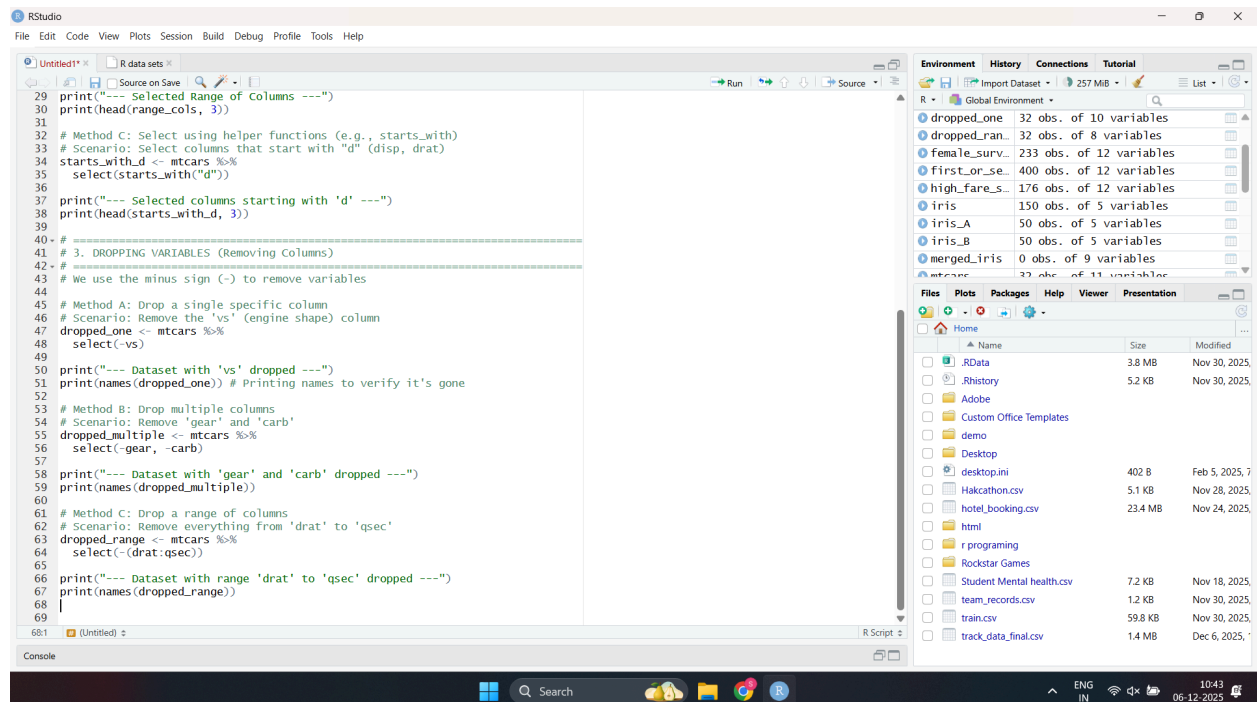
SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis SAS / SPSS / R

Aim - Selecting and dropping variables using select() in R. import dataset.

Input :



```
1 # 1. IMPORT DATASET
2
3 # library(dplyr) # select() is part of the dplyr package
4
5 # Use the built-in mtcars dataset
6 data("mtcars")
7
8 print("--- Original Dataset (First 3 rows) ---")
9 print(head(mtcars, 3))
10
11 # 2. SELECTING VARIABLES (Keeping Columns)
12
13 # Method A: Select specific columns by name
14 # Scenario: We only want Miles/(US) gallon (mpg), cylinders (cyl), and Horsepower (hp)
15 selected_cols <- mtcars %>%
16   select(mpg, cyl, hp)
17
18 print("--- Selected Specific Columns ---")
19 print(head(selected_cols, 3))
20
21 # Method B: Select a range of adjacent columns
22 # Scenario: Select everything from 'mpg' to 'disp'
23 range_cols <- mtcars %>%
24   select(mpg:disp)
25
26 print("--- Selected Range of Columns ---")
27 print(head(range_cols, 3))
28
29 # Method C: Select using helper functions (e.g., starts_with)
30 # Scenario: Select columns that start with "d" (disp, drat)
31 starts_with_d <- mtcars %>%
32   select(starts_with("d"))
33
34 print("--- Selected columns starting with 'd' ---")
35 print(head(starts_with_d, 3))
36
37 # 3. DROPPING VARIABLES (Removing Columns)
```



```
29 print("--- Selected Range of Columns ---")
30 print(head(range_cols, 3))
31
32 # Method C: Select using helper functions (e.g., starts_with)
33 # Scenario: Select columns that start with "d" (disp, drat)
34 starts_with_d <- mtcars %>%
35   select(starts_with("d"))
36
37 print("--- Selected columns starting with 'd' ---")
38 print(head(starts_with_d, 3))
39
40 # 3. DROPPING VARIABLES (Removing Columns)
41 # We use the minus sign (-) to remove variables
42
43 # Method A: Drop a single specific column
44 # Scenario: Remove the 'vs' (engine shape) column
45 dropped_one <- mtcars %>%
46   select(-vs)
47
48 print("--- Dataset with 'vs' dropped ---")
49 print(names(dropped_one)) # Printing names to verify it's gone
50
51 # Method B: Drop multiple columns
52 # Scenario: Remove 'gear' and 'carb'
53 dropped_multiple <- mtcars %>%
54   select(-gear, -carb)
55
56 print("--- Dataset with 'gear' and 'carb' dropped ---")
57 print(names(dropped_multiple))
58
59 # Method C: Drop a range of columns
60 # Scenario: Remove everything from 'drat' to 'qsec'
61 dropped_range <- mtcars %>%
62   select(-(drat:qsec))
63
64 print("--- Dataset with range 'drat' to 'qsec' dropped ---")
65 print(names(dropped_range))
66
67
68
69
```

Output :

Name - Siddhant Chalke
Roll No - S073

SUBJECT - Data Analysis SAS / SPSS / R

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for importing and manipulating the 'mtcars' dataset. The code includes comments explaining each step.
- Console/Terminal:** Shows the output of the executed commands, including variable names and data values.
- Environment Panel:** Lists loaded packages like 'dplyr' and 'tidyverse', and shows the contents of the global environment, such as 'dropped_one', 'female_surv', etc.

```
# =====>
# 1. IMPORT DATASET
# =====>
> library(dplyr) # select() is part of the dplyr package
> # Use the built-in mtcars dataset
> data("mtcars")
> print("--- Original Dataset (First 3 rows) ---")
[1] "--- Original Dataset (First 3 rows) ---"
> print(head(mtcars, 3))
      mpg   cyl  disp    hp  drat    wt    qsec vs am gear carb
Mazda RX4     21.0   6  160 110 3.90 2.620 16.46 0 1   4   4
Mazda RX4 Wag  21.0   6  160 110 3.90 2.875 17.02 0 1   4   4
Datsun 710     22.8   4  108  93 3.85 2.320 18.61 1 1   4   1
> # Method A: Select specific columns by name
> # Scenario: We only want Miles/(US) gallon (mpg), cylinders (cyl), and Horsepower (hp)
> selected_cols <- mtcars %>%
+   select(mpg, cyl, hp)
> print("--- Selected Specific Columns ---")
[1] "--- Selected Specific Columns ---"
> print(head(selected_cols, 3))
      mpg cyl  hp
Mazda RX4     21.0   6 110
Mazda RX4 Wag  21.0   6 110
Datsun 710     22.8   4  93
> # Method B: Select a range of adjacent columns
> # Scenario: Select everything from 'mpg' to 'disp'
> range_cols <- mtcars %>%
+   select(mpg:disp)
> print("--- Selected Range of Columns ---")
[1] "--- Selected Range of Columns ---"
> print(head(range_cols, 3))
      mpg cyl disp
Mazda RX4     21.0   6 160
Mazda RX4 Wag  21.0   6 160
Datsun 710     22.8   4 108
> # Method C: Select using helper functions (e.g., starts_with)
> # Scenario: Select columns that start with "d" (disp, drat)
> starts_with_d <- mtcars %>%
+   select(starts_with("d"))
> print("--- Selected columns starting with 'd' ---")
[1] "--- Selected columns starting with 'd' ---"
> print(head(starts_with_d, 3))
```

The screenshot displays the RStudio IDE interface. The main window shows a script editor with R code for data manipulation. The code includes comments and R commands to select, drop, and rename columns in a dataset named 'mtcars'.

```

# R 4.5.2 - ~/R
# Method C: Select using helper functions (e.g., starts_with)
# # Scenario: Select columns that start with "d" (disp, drat)
> starts_with_d <- mtcars %>%
+   select(starts_with("d"))
> print("--- Selected columns starting with 'd' ---")
[1] "--- Selected columns starting with 'd' ---"
> print(head(starts_with_d, 3))
      disp drat
Mazda RX4    160 3.90
Mazda RX4 Wag 160 3.90
Datsun 710   108 3.85

# # Method A: Drop a single specific column
# # Scenario: Remove the 'vs' (engine shape) column
> dropped_one <- mtcars %>%
+   select(-vs)
> print("--- Dataset with 'vs' dropped ---")
[1] "--- Dataset with 'vs' dropped ---"
> print(names(dropped_one)) # Printing names to verify it's gone
[1] "mpg" "cyl" "disp" "hp" "drat" "wt" "qsec" "am" "gear" "carb"

# # Method B: Drop multiple columns
# # Scenario: Remove 'gear' and 'carb'
> dropped_multiple <- mtcars %>%
+   select(-gear, -carb)
> print("--- Dataset with 'gear' and 'carb' dropped ---")
[1] "--- Dataset with 'gear' and 'carb' dropped ---"
> print(names(dropped_multiple))
[1] "mpg" "cyl" "disp" "hp" "drat" "wt" "qsec" "vs" "am"

# # Method C: Drop a range of columns
# # Scenario: Remove everything from 'drat' to 'qsec'
> dropped_range <- mtcars %>%
+   select(-(drat:qsec))
> print("--- Dataset with range 'drat' to 'qsec' dropped ---")
[1] "--- Dataset with range 'drat' to 'qsec' dropped ---"
> print(names(dropped_range))
[1] "mpg" "cyl" "disp" "hp" "vs" "am" "gear" "carb"

```

The right sidebar shows the 'Environment' pane with a list of objects and their sizes. The 'Files' pane shows the file explorer with a list of files and folders.

Object	Size	Modified
dropped_one	32 obs. of 10 variables	
dropped_range	32 obs. of 8 variables	
female_surv	233 obs. of 12 variables	
first_or_se	400 obs. of 12 variables	
high_fare_s	176 obs. of 12 variables	
iris	150 obs. of 5 variables	
iris.A	50 obs. of 5 variables	
iris.B	50 obs. of 5 variables	
merged_iris	0 obs. of 9 variables	
mtcars	32 obs. of 11 variables	

Name	Size	Modified
RData	3.8 MB	Nov 30, 2025
Rhistory	5.2 KB	Nov 30, 2025
Adobe		
Custom Office Templates		
demo		
Desktop		
desktop.ini	402 B	Feb 5, 2025, 7
Hakachthon.csv	5.1 KB	Nov 28, 2025
hotel_booking.csv	23.4 MB	Nov 24, 2025
html		
r programming		
Rockstar Games		
Student Mental Health.csv	7.2 KB	Nov 18, 2025
team_records.csv	1.2 KB	Nov 30, 2025
train.csv	59.8 MB	Nov 30, 2025
track_data_final.csv	1.4 MB	Dec 6, 2025

Name - Siddhant Chalke
Roll No - S073